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✓ Multiclass Classification using Logistic Regression :-

Assignment :-

Predict Employee Satisfaction Level (Low, Medium, High) based on:

Study Hours

Monthly_Salary Percentage

✓ Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

✓ Step 2: Create a Simple Multiclass Dataset

Target Variable: Classes:

0 → Low Satisfaction

1 → Medium Satisfaction

2 → High Satisfaction

```
data = {
    'Working_Hours': [
        35, 40, 45, 38, 42, 48, 50, 55, 60, 52,
        44, 58, 36, 41, 47, 49, 53, 57, 39, 43,
        46, 51, 54, 59, 37, 45, 50, 56, 48, 60
    ],
    'Monthly_Salary': [
        18000, 22000, 25000, 20000, 28000, 32000, 35000, 42000, 50000, 40000,
        30000, 48000, 19000, 24000, 29000, 33000, 38000, 45000, 21000, 27000,
        31000, 36000, 41000, 52000, 18500, 26000, 37000, 46000, 34000, 55000
    ],
    'Satisfaction': [
        0, 0, 0, 0, 1, 1, 2, 2, 2, 2,
        1, 2, 0, 0, 1, 1, 2, 2, 0, 1,
        1, 2, 2, 2, 0, 1, 2, 2, 1, 2
    ]
}

df = pd.DataFrame(data)
df
```

	Working_Hours	Monthly_Salary	Satisfaction
0	35	18000	0
1	40	22000	0
2	45	25000	0
3	38	20000	0
4	42	28000	1
5	48	32000	1
6	50	35000	2
7	55	42000	2
8	60	50000	2
9	52	40000	2
10	44	30000	1
11	58	48000	2
12	36	19000	0
13	41	24000	0
14	47	29000	1
15	49	33000	1
16	53	38000	2
17	57	45000	2
18	39	21000	0
19	43	27000	1
20	46	31000	1
21	51	36000	2
22	54	41000	2
23	59	52000	2
24	37	18500	0
25	45	26000	1
26	50	37000	2
27	56	46000	2
28	48	34000	1
29	60	55000	2

Step 3: Explore the Dataset

```
df.shape
```

```
(30, 3)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Working_Hours    30 non-null     int64
1   Monthly_Salary   30 non-null     int64
2   Satisfaction     30 non-null     int64
dtypes: int64(3)
memory usage: 852.0 bytes
```

```
df.describe()
```

	Working_Hours	Monthly_Salary	Satisfaction
count	30.000000	30.000000	30.000000
mean	47.933333	33416.666667	1.166667
std	7.487614	10612.357252	0.833908
min	35.000000	18000.000000	0.000000
25%	42.250000	25250.000000	0.250000
50%	48.000000	32500.000000	1.000000
75%	53.750000	40750.000000	2.000000
max	60.000000	55000.000000	2.000000

Step 4: Split Features and Target

```
X = df[['Working_Hours', 'Monthly_Salary']]
y = df['Satisfaction']
```

Step 5: Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Step 6: Create and Train Logistic Regression Model

```
model = LogisticRegression(multi_class='multinomial', solver='lbfgs', max_iter=1000)
model.fit(X_train, y_train)
```

/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning: 'multi_class' was deprecated in v

```
LogisticRegression
LogisticRegression(max_iter=1000, multi_class='multinomial')
```

Step 7: Model Prediction and Evaluation

```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
Classification Report:
              precision    recall  f1-score   support

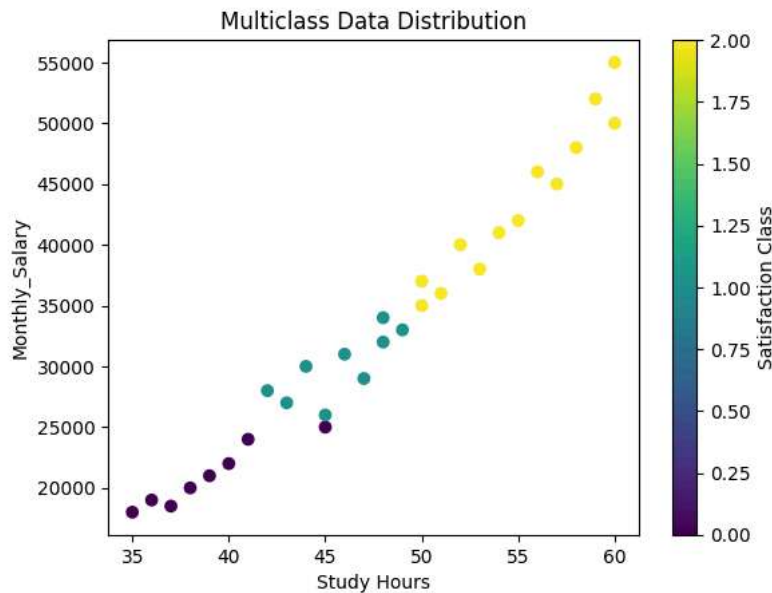
     1         1.00      1.00      1.00         1
     2         1.00      1.00      1.00         5

   accuracy          1.00
  macro avg          1.00
 weighted avg          1.00
```

Step 8: Visualization

```
plt.scatter(df['Working_Hours'], df['Monthly_Salary'], c=df['Satisfaction'], cmap='viridis')
plt.xlabel('Study Hours')
plt.ylabel('Monthly_Salary')
plt.title('Multiclass Data Distribution')
```

```
plt.colorbar(label='Satisfaction Class')
plt.show()
```



Step 9: Prediction Using User Input

```
Working_Hours = float(input("Enter Working hours: "))
Monthly_Salary = float(input("Enter Monthly_Salary percentage: "))
```

```
new_data = [[Working_Hours, Monthly_Salary]]
prediction = model.predict(new_data)
```

```
labels = {0: 'Low', 1: 'Medium', 2: 'High'}
print("Predicted Satisfaction Level:", labels[prediction[0]])
```

```
Enter Working hours: 50
Enter Monthly_Salary percentage: 40000
Predicted Satisfaction Level: High
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
warnings.warn(
```

```
import pickle
```

```
pickle.dump(model, open('MModel.pkl', 'wb'))
```

Flask Code

Save this file with 'app.py'

```
from flask import Flask, render_template, request
import pickle

app = Flask(__name__)

# Load Trained Model
model = pickle.load(open("MModel.pkl", "rb"))

@app.route('/')
def home():
    return render_template("index.html")

@app.route('/predict', methods=['POST'])
def predict():

    # Read input values
    working_hours = float(request.form['working_hours'])
```

```

monthly_salary = float(request.form['monthly_salary'])

# Prediction
prediction = model.predict([[working_hours, monthly_salary]])

# Convert class number into satisfaction level
if prediction[0] == 0:
    satisfaction = "Low Satisfaction"

elif prediction[0] == 1:
    satisfaction = "Medium Satisfaction"

else:
    satisfaction = "High Satisfaction"

return render_template(
    "index.html",
    prediction_text=f"Predicted Job Satisfaction : {satisfaction}"
)

if __name__ == "__main__":
    app.run(debug=True)

```

* Serving Flask app '__main__'
 * Debug mode: on
 INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server inst
 * Running on <http://127.0.0.1:5000>
 INFO:werkzeug:Press CTRL+C to quit
 INFO:werkzeug: * Restarting with watchdog (inotify)

HTML Code

Save this file with 'index.html'

```

<!DOCTYPE html>
<html lang="en">

<head>

<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Employee Satisfaction Predictor</title>

<style>

*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial, Helvetica, sans-serif;
}

body{
  background:#eef5ff;
  display:flex;
  justify-content:center;
  align-items:center;
  min-height:100vh;
}

.container{
  width:480px;
  background:white;
  padding:35px;
  border-radius:15px;
  box-shadow:0px 10px 25px rgba(0,0,0,.2);
}

h1{
  text-align:center;
  color:#1565c0;
}

```

```

h3{
  text-align:center;
  color:#555;
  margin-top:8px;
  margin-bottom:25px;
}

label{
  font-weight:bold;
  display:block;
  margin-top:15px;
  margin-bottom:6px;
}

input{
  width:100%;
  padding:12px;
  border:1px solid #ccc;
  border-radius:8px;
  font-size:16px;
}

button{
  width:100%;
  margin-top:25px;
  padding:13px;
  background:#1565c0;
  color:white;
  border:none;
  border-radius:8px;
  font-size:18px;
  cursor:pointer;
  transition:.3s;
}

button:hover{
  background:#0d47a1;
}

.result{
  margin-top:25px;
  background:#e8f5e9;
  padding:18px;
  border-radius:8px;
  text-align:center;
  color:green;
  font-size:21px;
  font-weight:bold;
}

footer{
  text-align:center;
  margin-top:20px;
  color:#777;
  font-size:13px;
}

</style>

</head>

<body>

<div class="container">

<h1>Employee Satisfaction Predictor</h1>

<h3>Multiclass Classification using Logistic Regression</h3>

<form action="/predict" method="POST">

<label>Working Hours per Week</label>

<input
type="number"
name="working_hours"
step="1"
min="1"

```

```
max="100"
placeholder="Enter Working Hours"
required>

<label>Monthly Salary (₹)</label>

<input
type="number"
name="monthly_salary"
step="500"
min="5000"
placeholder="Enter Monthly Salary"
required>

<button type="submit">
Predict Satisfaction
</button>

</form>

{% if prediction_text %}

<div class="result">
{{ prediction_text }}
</div>

{% endif %}

<footer>
Employee Job Satisfaction Prediction
</footer>

</div>

</body>

</html>
```